Nash Equilibria of the Cournot Game through the lens of the Debreu-Fan-Glicksberg Theorem

Consider this Cournot game with $n \geq 2$ firms. The inverse demand and each firm's cost function satisfy the conditions of the textbook monopoly model I introduced earlier. Namely, the inverse demand $p(\cdot) : \mathbb{R}_+ \to \mathbb{R}_+$ is continuous and nonincreasing; and on $\{Q \geq 0 \,|\, p(Q) > 0\}$, it is C^1 with $p' < 0$; and each firm i's cost function $c_i : \mathbb{R}_+ \to \mathbb{R}_+$ is C^1 with $c_i(0) = 0$ and $c_i' > 0$. In addition to these assumptions there is a $\overline{Q} > 0$ such that $p(Q) > 0$ if and only if $0 \leq Q < \overline{Q}$. Without loss of generality, we may take each firm i's strategy set to be $S_i = [0, \overline{Q}]$, since any $q_i > \overline{Q}$ is strictly dominated by $q_i = 0$.

With this compactification, each firm's strategy set is nonempty, compact, and convex. Firm i's profit is $u_i(q_1, ..., q_n) = p(\sum_j q_j)q_i - c_i(q_i)$, which is continuous in $q = (q_1, ..., q_n)$ on $S = [0, \overline{Q}]^n$.

The only remaining condition to check is that each firm's profit is quasiconcave in its own output on $[0, \overline{Q}]$. Suppose in addition each c_i is *convex* and the inverse demand is *concave* on $[0, \overline{Q}]$. First it is easy to see that each $u_i(\cdot, q_{-i})$ is *concave* on $[0, \overline{Q} - \sum_{j \neq i} q_j]$. Cost is convex on $\mathbb{R}_+$ and the derivative of firm i's revenue is

$$\frac{\partial}{\partial q_i} p(Q)q_i = p'(Q)q_i + p(Q).$$

on $[0, \overline{Q} - \sum_{j \neq i} q_j]$ and 0 otherwise. On this interval, firm i's marginal revenue is decreasing and so revenue is concave on this interval. Unfortunately, revenue cannot be concave in q_i on $[0, \overline{Q}]$, since it must be zero whenever $Q > \overline{Q}$, and if $q_j > 0$ for some $j \neq i$, some values of q_i in $[0, \overline{Q}]$ imply that $Q > \overline{Q}$. So we cannot hope to show that $u_i(\cdot, q_{-i})$ is concave on $S_i = [0, \overline{Q}]$.

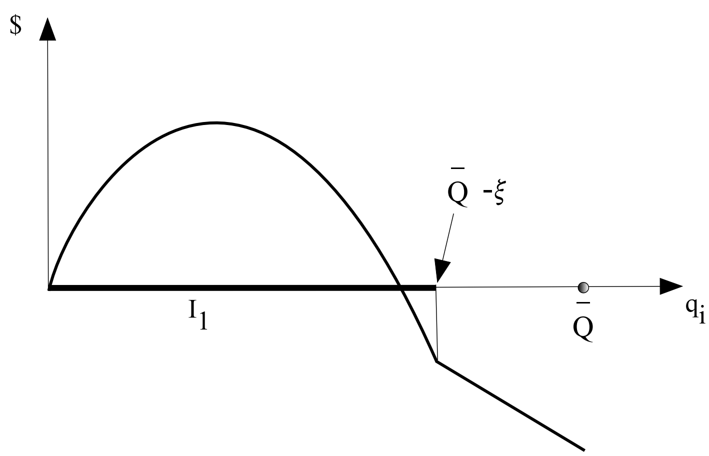

Figure 1: Firm i's profit on $S_i = [0, \bar{Q}]$. On the interval I_1, profit is concave. Here firm i's cost is linear, as reflected in the straight-line segment from $\bar{Q} - \xi$ to $\bar{Q}$. Had firm i's cost been strictly convex, then profit would be strictly concave on that interval, but there would still be a kink; in that case you can view the straight line as the tangent line to firm i's profit. Firm i's profit cannot be concave in q_i, but it is quasiconcave.

Nonetheless it turns out that $u_i(\cdot, q_{-i})$ is *quasiconcave* on $[0, \overline{Q}]$. Of course the function $-c_i(\cdot)$ is concave; and revenue $p(q)q_i$ is quasiconcave, since revenue is concave on $[0, \overline{Q} - \sum_{j\neq i} q_j]$—the interval on which price and revenue is positive—and zero otherwise. So profit is the sum of quasiconcave functions. Unfortunately the sum of quasiconcave functions is not necessarily quasiconcave.[1] But all is far from lost, since we can just apply the definition of quasiconcavity directly to $u_i(\cdot, q_{-i})$. Let $\xi = \sum_{j\neq i} q_j$, the sum of output of other firms, which I hold fixed here. If $\xi \geq \bar{Q}$ then the firm's profit is $-c_i(q)$, which concave on S_i; if $\xi = 0$, then firm i's profit is $p(q_i)q_i - c_i(q_i)$, which is concave on S_i. So suppose that $0 < \xi < \bar{Q}$. Write firm i's profit on $S_i = [0, \overline{Q}]$ as

$$f(q_i) = p(q_i + \xi) - c_i(q_i)$$

On $I_1 = [0, \overline{Q} - \xi]$ and on $I_2 = [\overline{Q} - \xi, \overline{Q}]$, f is concave, being the sum of concave functions. The only way that the definition of quasiconcavity might not hold is by comparing outputs across these two intervals.[2] Let $q_i^t \in I_t$, and let $q_i^\lambda = \lambda q_i'' + (1-\lambda)q_i'$ for $0 \leq \lambda \leq 1$. Note that $f(q_1^1) \geq f(q_1^2)$, since revenue is no lower and cost no higher for q_1^1 than for q_1^2. If $q_1^\lambda \in I_1$, then $f(q_1^\lambda) \geq f(q_1^2)$ by the just-mentioned fact; and if $q_1^\lambda \in I_2$, then $f(q_1^\lambda) \geq f(q_1^2)$ since cost is increasing. So f is quasiconcave on S_i.

I used concavity of $p(\cdot)$ on $[0, \overline{Q}]$ to show that each firm's revenue is concave on $[0, \overline{Q} - \xi]$. Concavity of $p(\cdot)$ can be replaced with "$p(\cdot)$ is *not too convex* in the Arrow-Pratt sense." More precisely, you can verify that each firm i's revenue is concave on $[0, \overline{Q} - \xi]$ if $p(\cdot)$ is *more concave than* $f(Q) = Q^{-1}$ on $[0, \overline{Q}]$ in the sense that $p = T \circ f$, for some concave, strictly increasing T. Since f is a strictly convex function, $p(\cdot)$ need not be concave.

[1] For example the functions $f(x) = -x$ and $g(x) = x^2$ on $\mathbb{R}_+$ are monotone, and so each is quasiconcave. But $h(x) = f(x) + g(x) = -x + x^2$ is not quasiconcave.

[2] Alternatively, you can show that f is single-peaked; since any single-peaked function on a real interval is quasiconcave, the conclusion follows.